# Two maps on the circle are pointwise equal as soon as they agree at the base point and on the loop

The Univalent Foundations Program

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*Editorial note.* The counted statement is the author lem environment at source lines 309-318 and its proof environment opens on line 319, the next line; the pairing is by adjacency and the retained proof range 319-325 is the whole environment.

*Editorial note.* The pinned book label-number dump carries no entry for this label, so no counter value is imposed and the counted lemma prints with the wrapper numbering; the original number is not claimed.

*Editorial note.* Two labels the proof cites lie outside this chapter and are declared in external_references as non-load-bearing navigation pointers: thm:transport-path and thm:dpath-path, both in the basics chapter; each is counted by its own card of this lane and the retained proof only uses them to reduce the loop case.

*Editorial note.* The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

*Editorial scope.* One original result of the pinned HoTT Book source tree, counted once: the uniqueness principle for maps on the circle (author label thm:uniqueness-for-functions-on-S1).

*Allowed standard prerequisites (editorial).*

- The induction principle of the circle. To prove a family over the circle it suffices to give an element over the base point together with a path over the loop between its transports.. The whole proof is that induction applied to the family of equalities between the two maps.
- Transport in path spaces over a loop. The transport of a path between the two values at the base point along the loop, inside the family of equality types, reduces to the given comparison between the two actions on the loop.. It is the reduction that turns the loop case into the datum q of the statement.

*Source reference.* Original theorem 2.11.3: sources/hott-basics/pinned-original-source.tex, source lines 1896–1896. Book Theorem 2.11.3, on transport in a family of path types, used in the retained proof to reduce the loop case of the

circle induction. Navigation pointer only: it is counted by its own card of this lane.

*Source reference.* Original theorem 2.11.5: sources/hott-basics/pinned-original-source.tex, source lines 1919–1919. Book Theorem 2.11.5, identifying dependent paths with ordinary paths, used together with the previous result to turn the loop case into the given comparison. Navigation pointer only: it is counted by its own card of this lane.

**Lemma 0.0.1.** *If $A$ is a type and $f, g : \mathbb{S}^1 \to A$ are two maps together with two equalities $p, q$:*

$$p : f(\mathsf{base}) =_A g(\mathsf{base}),$$
$$q : f(\mathsf{loop}) =_p^{\lambda x.\, x =_A x} g(\mathsf{loop}).$$

*Then for all $x : \mathbb{S}^1$ we have $f(x) = g(x)$.*

*Proof.* We apply the induction principle of $\mathbb{S}^1$ at the type family $P(x) :\equiv (f(x) = g(x))$. When $x$ is $\mathsf{base}$, $p$ is exactly what we need. And when $x$ varies along $\mathsf{loop}$, we need $p =_{\mathsf{loop}}^{\lambda x.\, f(x)=g(x)} p$, which by original theorems 2.11.3 and 2.11.5 can be reduced to $q$. □